

NeuroCA: A Hybrid Neural Network Based on Cellular Automata

From Register Neurons to a Verifiable Code Generator — Series v0.1-v120

NeuroCA Laboratory

Independent research / DeepSeek Harness lab

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Reproducibility-first preprint. All artifacts (checkpoints, corpora, reports, registry) are versioned and referenced in §8. The evaluation harness is exec-based (compile + real tests in a sandbox); the format guard and the Tier1 audit are described in §4.3 and §6.4.

Abstract

We present NeuroCA, a research neural-network architecture in which a neuron is interpreted as a W -bit register on an N -dimensional torus, and the whole network as a binary cellular automaton (CA) in $N+1$ dimensions. Unlike classical neural networks that train millions of weights "blindly", NeuroCA keeps the main computational dynamics fixed and local (given by cellular-automaton rules), while only a compact readout layer over deterministic reservoir features is trained. Across the version series v0.1-v120 the architecture evolved from a single 4D machine (5832 bits) through agent networks, rule evolution, STaR self-training and a dual-loop "generate + assemble" system, to a hybrid of a register CA substrate (~530 features) and a Transformer/MoE (53M parameters) solving 33 algorithmic tasks with a full exec-run of $658/660 = 99.7\%$ (etalon v120big50_kit2b).

Key measured results and lessons of the series: (1) model capacity matters — growing parameters from 7.2M to 53M removes the "binarity" of training profiles (feature overlap $0.764 \rightarrow 0.285$); (2) quality growth comes from data, not parameters ($544 \rightarrow 648 \rightarrow 658$ as the corpus expands; epoch plateau); (3) in a controlled ablation at the current scale the CA substrate adds $+5.0 \pm 5.8$ p.p. (within noise), while data has a $\sim 18\times$ larger effect; (4) synthetic elisions/template steps in the corpus hurt (base-only > synthetic); (5) the LLM-teacher loop (qwen2.5:14b, yield $\sim 51\%$ EXEC-valid) works: a real defect in second_max was fixed $0 \rightarrow 20/20$; (6) the distillation pilot with an external LLM achieved 9/10 on the first attempt; (6a) the decisive edge-data test (kit2c, 1161 rows) refuted the hypothesis "Tier1 zeros = uncovered inputs": edges gave no gain ($660 = 656$, Tier1 485 vs 487 for kit2b) — the next lever is staged loss decomposition; (7) "the harness can lie": the eval-test format bug understated the result (82.4% instead of the real 99.8%), and a Tier1 evaluation defect (overwrite of a repeated family) makes the 640 denominator ambiguous — a metric is only as honest as the harness.

We discuss limitations (closed vocabulary $V=1078$, narrow base tests, no functional tests for Stack/Queue, Tier1 failures), and directions: staged loss decomposition, experts $8 \rightarrow 16$, Tier2 compositions via distillation, and a controlled substrate ablation at a $\times 5$ data scale.

Keywords: cellular automata, register machine, reservoir computing, code generation, STaR self-training, LLM teacher, distillation, Mixture-of-Experts,

exec-verification, honest metrics.

1. Introduction

1.1 Motivation

A classical neural network trains millions of weights and remains a "black box": its state is not observable and result verification is heuristic. NeuroCA proposes an alternative: train a thin readout layer, while the main computational load is carried by the deterministic local dynamics of a cellular automaton, where each neuron is a register of bits, and the rules are local, interpretable, and not trained.

The motivation is threefold:

1. Interpretability — the machine state is a bit array; it can be observed, edited, and verified.
2. Robustness — local rules provide self-repair: a damaged state is restored by the dynamics.
3. Honest verification — in the applied setting (code generation) the result is not assessed subjectively but executed: compile + real tests in a sandbox.

1.2 Problem statement and working hypothesis

Task: given a Russian-language formulation of an algorithmic task (e.g., "задача find_max" — "task find_max"), generate valid Python code that passes compilation and functional exec on tests.

Working hypothesis: the dynamics of a cellular automaton over registers can replace most of the trained capacity, leaving to learning only the "reading" of reservoir features. The final metric is the share of generations that are actually executable and correct on tests (EXEC).

1.3 Contributions

1. The architecture "neuron = register, network = CA in $N+1$ dimensions" with an open set of local bit rules (majority, Toom, HDC/VSA, Life, Kauffman, identity).
2. The hybrid scheme HybridPrefix + Substrate3: a trainable Transformer/MoE over a deterministic CA reservoir (~530 features), with CA-aware gating.
3. A complete honest-CI loop: version registry, format guard, 660 exec-protocol, regression gate, article auto-sync (plugin neuroca-lab).
4. An LLM-teacher loop (an external LLM as a "more knowledgeable other") and a distillation pilot of a verified corpus.
5. Systematic recording of negative results (synthetic data, LoRA, head blocks, frozen core, hybrid profiles) as scientifically valuable data.

2. Related Work

Cellular automata as computational models were systematized by Wolfram [1]. Growing Neural Cellular Automata demonstrated self-repair and morphogenesis [2]. Reservoir computing uses fixed nonlinear dynamics with a trainable readout [3]; NeuroCA inherits this idea, replacing the analog reservoir with a bit-level CA. Self-taught reasoner (STaR) [4] is the basis of the self-correction loop; language model scaling laws [5, 6] and emergent abilities [7] frame the role of capacity (measured here as G1: feature overlap 0.285 at 53M). Mixtures of agents [8] and MoE layers [9] are the architectural basis for expert expansion. Lindenmayer L-systems [10] and cognitive maps (O'Keefe, Moser; Tolman) [11, 12] motivate the "living text" and spatial-understanding directions. Interiorization (Vygotsky) [13], dialogism (Bakhtin) [14], predictive coding (Friston) [15], and dual-process theory

(Kahneman) [16] provide the theoretical frame for an internal-dialogue analog. LoRA [17] was used in negative experiments of modular growth.

3. Methods

3.1 RegisterCA core: neuron = register

The machine state is a uint8 array of shape spatial_shape + (W,). A cell of the torus lattice holds a W-bit register; bit b of a cell interacts with spatial neighbors (same bit) and with register neighbors $b \pm 1$ of the same cell — the register becomes the (N+1)-th axis of the machine. The flagship configuration: a $9 \times 9 \times 9$ torus $\times$ 8 bits = 5832 bits.

Core API: reset(zeros|random), set_bitplane (input injection), step(n, noise_p, async_frac) (synchronous/asynchronous dynamics with noise), features(bits|grid|flat), bitplane_densities, popcount. Throughput is ~130 million bit-updates/s on CPU without bit-packing.

Measured fact "register as an axis": without bit-plane coupling the signal does not spread across the register; coupling (Toom/HDC rules) genuinely transfers information between bits of a cell (v0.1 experiments: 4D-machine dynamics gives 0.929 vs 0.844 for static injection, +8.5 p.p.).

3.2 Local dynamics rules

All rules are purely bit-level; $\text{MAJ3}(a,b,c) = (a \& b) | (a \& c) | (b \& c)$.

Rule | Mechanics

majority_nd | majority of $2N+1$ votes (N — torus dimension)
toom_reg | $\text{MAJ}(\text{axis-0 neighbor, center, register bit } b+1)$ — Toom's rule
hdc | $R \leftarrow \text{MAJ}(R, \text{rot}_k(R(+e_x)), \text{rot}_m(R(+e_y)))$ — HDC/VSA binding

life_reg | Life in N+1 dimensions

kauffman | **K=2 random inputs + own LUT (subject of evolution)**

identity | **identity (delay-line, baseline)**

3.3 Substrate3 features (CA reservoir)

Substrate3 is a deterministic set of ~530 CA-state features computed at each decoding step: a shift chain of the last 12 tokens, syntax detectors, ripple-carry counters, phase clocks, block counters 8/16/32/64, conjunctions, and RBN subnets. Features are computed in a GPU cache and concatenated with the transformer state at each token. The substrate is not trained and not partitioned (shared memory = the abstraction mechanism; probe qtree 0.826).

3.4 HybridPrefix architecture

task (Russian text)
| tokenization (reversible, round-trip 100%, V=1078)

▼

HybridPrefix:

tok + pos → L×[Attention + MoE(8 experts, top-3) + Substrate gating]

d=256, L=6, H=8, FDIM=530, R=2

→ head(torch.cat([h, last_feats]))

| cascade: <A> analysis → <S> strategy → <C> scheme → <K> code

▼

Python code → compile → EXEC on tests (sandbox)

Parameters: Branch A — 53M (d=256, L=6, H=8, MoE 8/3, R=2); early versions — 7.2M (V=1078). The entire transformer is trained (not a "thin layer"). The cascade agent "task → analysis → strategy → scheme → code": early stages use an internal "pidgin" (in-vocab), code is the target Python.

3.5 Training

- Optimization: AdamW (lr=3e-4 for scratch, **lr=1e-4 for fine-tuning from a checkpoint — 3e-4 destabilizes: ep112 76→36), weight_decay 1e-4, cosine annealing, bf16 autocast + GradScaler, grad clip 1.0, batch 160-256.
- Training on the GPU feature cache; data are (prefix, next-token) pairs from labels/*.jsonl / .npz corpora (up to 153-155K pairs).
- STaR self-training: generate → filter (compile+exec) → fine-tune on the mixture; anti-saturation (skeleton cap, double weight for new examples) unlocks rounds 2+.
- Full training v0.59-v0.61: 0.98M parameters, WIN=150; records rate 69.27%, E 39.72% before corpus expansion.

3.6 LLM-teacher loop and distillation

Teacher: Ollama qwen2.5:14b (semantics), qwen2.5-coder:14b / deepseek-coder (code), backup minimax-m3:cloud (pilot G6: 9/10 first-try, latency 0.7-2.5s).
Prompt: "task, analysis, strategy, scheme, code, tests"; double filter (vocabulary parser + compile + exec) → only verified rows enter the corpus.

Measurements: 208 generations → 106 EXEC-valid (~51% yield) in ~30 min; a real defect in second_max (0/20 in v114b) was fixed to 20/20; average 15→19/20.
Batch generation of the full cascade (B=4) is ×2.3 faster and better than stage-by-stage (fixed bubble_sort).

Distillation (path "b"): NeuroCA 53M = planner (task → in-vocab analysis/strategy) + LLM-generator (general code) + verifier (exec) + retry loop; verified rows are distilled into the student corpus. This removes the closed vocabulary blocker for Tier2 compositions (the teacher reads arbitrary names).

4. Experimental Setup

4.1 Corpora

The corpus grew: 91 → 419 → 519 examples (v112-v114c), then expansion to 33 families (v115+): 2094 rows (kit2: base 1054 / A-elision 522 / R-steps 483 / weak_fix 35), kit2_edges +72 edge rows (24 families × 3 variants), Branch A — 153-155K pairs (v121: 1532 rows, weight 3-5, UNK=0). Decision: synthetic (elisions/template steps) is excluded from the corpus — base-only > synthetic.

4.2 Benchmarks

Bench | Composition | What it measures

660 (base) | 33 families × 20 samples | generation stability on base tests

Tier1 | 32 bench_tier1 items (31 unique families) × 20 | behavior on new inputs (edges: []→None, 2^-1, -123, tabs, -1234/10^9) with the same prompts

Tier2 | 12 v122 compositions × 5-20 | skill compositions (sort_and_search etc.) — out-of-corpus

Measured: 660 = 658/660 (99.7%), Tier1 = 487 over saved entries (31×20 = 620; the declared 640 denominator is ambiguous — audit §6.4), Tier2 = 0/60 (OOV-name blocker, §6.5).

4.3 Eval protocol and format guard

1. Format guard (mandatory): tests of the form ((args...), expected), NOT (a, b, expected) — a historical eval bug since v112 "killed" all multi-argument functions (artifact: 82.4% instead of the real 99.8%).

2. 660 run through the 16-thread CPU pool (exec harness, ~1200-1500 s), exec cache by code hash.
3. Per-sample accounting: {exec, compile, name}; the main metric is exec.
4. Auto-detection of log encodings (UTF-8/UTF-16/binary).

4.4 Gates and promotion criteria

Gate | Condition | Result

G0 | Tier2 EXEC 100% | PASSED (data ready)

G1 | 50M overlap < 0.3 | PASSED (0.285) → Branch A

G2 | $660 \geq 651$, Tier1 ≥ 529 , Tier2 $\geq 30\%$ | not strictly met (Tier2 not measurable)

G6 | distillation pilot 9/10 | PASSED

Regression gate | pct drop ≥ 2 p.p. OR any task $\geq 20\%$ | FAIL for v114d, v116, v117e etc.

STOP-REGRESSION | -14 on 8 seeds | auto-rollback to old peak

Promotion to etalon requires a PASS gate and human confirmation (auto-promotion is forbidden; lesson v114d).

5. Results

5.1 Series timeline (key milestones)

Version(s) | Method | Key result

v0.1 | 4D machine $9 \times 9 \times 9 \times 8$ | digits 0.929 (+8.5 p.p. by dynamics); 130M bit/s

v0.2-0.3 | agents + hierarchy | routing 1.000; hierarchy 0.564 vs 0.262

v0.6-0.11 | ES rule evolution | honest negative; diversity = model property

v0.24-0.30 | sliding window + linear | ppl 2.00; reservoir saturation; $P \sim \text{ppl}^{-\text{len}}$

v0.34-0.36 | TinyGPT vocab500 + mono-detok | 42% (was 0.13%) → 62.1% with names+F1

v0.39 | STaR | 73.7% (+12.3 p.p.)

v0.42-0.48 | substrate + MLP → hybrid | 20.6% → 57.1% ($\times 440$ vs v0.28)

v0.49-0.54 | STaR r2 + assembly + critic | 65.09%; best-of-4 97.62%; E 99.75%, rate $\times$ E 0.995

v0.59-0.61 | full training 0.98M | rate 69.27% — record; E 39.72%

v112-v114c | +teacher loop, honest eval | EXEC 459/460 = 99.8% (after format fix)

v115-v117 | expansion to 33 families | 583 → 628/660; second_max — problem family

v118_sub / A2 | substrate ablation | $+5.0 \pm 5.8$ (noise); data $\sim 18\times$ more important

v119-v120 | canonization, warmup, replay | remove_spaces 0→20; Tier1 record 529 (replay2)

v120big50_v121 | Branch A 53M, 153K pairs | 648/660 (old 450, new 198), Tier1 474

v120big50_kit2b | Branch A, base-only 1089 rows | 658/660 = 99.7%, Tier1 487 — etalon

5.2 Version registry (26 entries, etalon ★)

| Version | Verdict | EXEC/total | pct | Note

1 | v114c | in_progress | 459/460 | 99.8% | historical etalon, 23 tasks

2 | v114d | rejected | 435/460 | 94.6% | -5.2 p.p.; average 19→0

3 | v115 | in_progress | 583/660 | 88.3% | expansion to 33 families

4 | v116 | rejected | 551/660 | 83.5% | -10.4; power 20→6

5 | v116b | rejected | 577/660 | 87.4% | -5.0; second_max 20→0

6 | v116c | under_review | 613/660 | 92.9% | -0.2; second_max 20→18

7 | v117 | under_review | 628/660 | 95.2% | -0.4; second_max 20→17

8 | v117e | rejected | 589/660 | 89.2% | -6.7; remove_spaces 20→3

9 | v118_sub | experiment_closed | 587/660 | 88.9% | ablation: $+5.0 \pm 5.8$ (noise)

10 | v118_a2control | control | 497/660 | 75.3% | no seeding: merge_sorted 0

11 | v119transfer | under_review | 618/660 | 93.6% | UNK canonization: remove_spaces 0→20

12 | v119scratch | control | 527/660 | 75.3% | canonization w/o transfer: +40 SWA

13 | v119replay | control | 541/660 | 82.0% | Tier1 SWA 502 — series record

14 | v120frozen_blocks | control | 621/660 | 94.1% | frozen core + head blocks: old 458

15 | v120warmup | control | 625/660 | 94.7% | warmup base-only: caesar 20→15

16 | v120warmup2 | rejected (replaced) | 651/660 | 98.6% | base-profile balance champion

17 | v120replay2 | control | 596/660 | 90.3% | Tier1 record 529/640

18 | v120replay3 | control | 596/660 | 90.3% | determinism confirmation
 19 | v120hybrid | control | 650/660 | 98.5% | base/edge compromise not confirmed
 20 | v120edgeblock | control | 264/264 (check) | 100% | edge head does not specialize
 21 | v120lora_e1 | control | 264→109 | — | LoRA edge-only destroys base
 22 | v120lora | control | 264→199 | — | LoRA 50/50 too
 23 | v120big50 | G1-PASSED | 114/264 (check) | 43% | overlap 0.285 → Branch A
 24 | v120big50_v121 | Branch A final | 648/660 | 98.2% | new 198 > etalon 192; Tier2 0/60
 25 | ★ v120big50_kit2b | etalon | 658/660 | 99.7% | base-only 1089 rows; Tier1 487
 26 | v120big50_kit2c | final (etalon unchanged) | 656/660 | 99.4% | base+weak_fix+edges (1161 rows); old 460, new 196, Tier1 485 — edges gave no gain (noise ±2)

5.3 Per-task breakdown of the etalon (660)

All 33 families $\geq 18/20$; the only base failure is is_balanced_parentheses (2 of 20 samples). Old 23 families: 460/460 (100%). New 10: 198/200 (99%).
 Tier1 weak spots (0/20 while 660 = 20/20): find_max, power, digit_sum, remove_spaces, count_digits — interpretation: uncovered edge inputs (negative numbers, []→None, tabs, 0^0 , 2^{-1}), not rephrasings.

5.4 Negative results (scientifically valuable)

1. Synthetic data (A-elision/R-steps) in the corpus — v121warmupA1: retrieval 19%→15%, 660 −8; kit2b A/B: base-only (1089) > synthetic (1532) at 53M.
2. LoRA channels over a frozen core — v120lora_e1 264→109, v120lora 264→199: corrections dominate the frozen core without a router.
3. Head-only blocks over a frozen core — v120frozen_blocks 621 (new 163 = v117), v120edgeblock (edge head overfits: Tier1 49→21): heads cannot carry specialization; depth is required.
4. Hybrid profile (warmup + edge) — v120hybrid 650/454: no compromise between the base champion (651/477) and the edge champion (596/529) exists; the space is binary until capacity scaling.
5. Substrate at current scale — v118_sub: $+5.0 \pm 5.8$ p.p. (within noise).

5.5 Ablation: data vs substrate vs capacity

- Control A2 (scratch without seeding, 497/660) vs v118_sub (586.7): a 89.7 difference — the effect of data, not architecture; factor $\sim 18\times$.
- G1 probe: feature overlap 0.764 → 0.285 at 7.2M→53M (ridge_sep 0.842, MLP_sep 0.9): capacity removes binarity.
- 53M: 94K pairs → 544; 153K pairs → 648; base-only 1089 rows → 658: **growth from data, epoch plateau (25 vs 35).

6. Discussion

6.1 CA as extended mind

Measurements v118–v120 show: the CA substrate in late NeuroCA is not a "thinking part" but a deterministic provider of features (extended mind; probe qtree 0.826). Its contribution at the current scale is small ($+5.0 \pm 5.8$), but this does not prove the CA is useless: the narrow benchmark (33 iterative algorithms) and small data do not let the reservoir show compositional compression. Open hypothesis: the substrate's contribution grows with data scale and task compositionality (Tier2). Verification — a controlled with/without-substrate ablation at a $\times 5$ data scale, 3 seeds.

6.1a Negative result of kit2c: edge data do not cure Tier1

The decisive experiment of the hypothesis "Tier1 zeros = uncovered inputs" is complete: kit2c (base 1054 + weak_fix 35 + edges 72, 1161 rows, transfer from kit2b-best) gave 660 = 656 (old 460, new 196), Tier1 = 485 — below kit2b

(658/487). Edge rows (empty lists []→None, tabs, 0^0, negative numbers) already present in the kit2c corpus gave no gain (noise ± 2). Consequence: Tier1 failures are not simply uncovered inputs but a deeper problem of mapping "formulation/inputs → family → strategy"; the next lever is staged loss decomposition (task→analysis→strategy→code) and/or formulation distillation via the teacher, not more edge data.

6.2 Data versus parameters

Twice confirmed: quality grows from data (544→648→658), not from epochs/size (plateau 25 vs 35 epochs at 53M). Hence the order "annotation → data → parameters"; the RTX 5080 parameter ceiling (~70–80M) is not the bottleneck before corpus expansion.

6.3 Profile binarity and capacity

Before Branch A, profiles were "binary": either the base champion (651/477) or the edge champion (596/529), no compromise possible. G1 (overlap 0.285 at 53M) showed the cause is insufficient capacity, not a fatal incompatibility: with more capacity the model masters both profiles (kit2b: 658/487).

6.4 Harness honesty (metrics)

Two confirmed evaluation defects:

1. Test format (a,b,expected) instead of ((args...),expected) since v112 understated the result: 82.4% instead of the real 99.8% (fixed by the format guard).
2. Tier1 audit (26.09): res[fam]=got overwrites the result of a repeated family (decimal_to_binary appears twice in the bench) — $31 \times 20 = 620$ saved, not 640; t1_Stack/t1_Queue are absent from the report; 660-Stack/Queue are scored by compile+name without functional tests (tests=None).

Conclusion: a metric is only as honest as the harness; the P0 priority is fixing eval (dedup, denominator by fact, per-sample, functional Stack/Queue tests, hashes of bench/model/corpus) before final measurements.

6.5 Limitations

1. Closed vocabulary V=1078: Tier2 composition names (sort_and_search etc.) are OOV; text_to_tokens drops them → Tier2 = 0/60 is not measurable without V expansion or teacher-based distillation (path "b").
2. Narrow base tests 660: is_sorted (only sorted), is_leap (only 2000), second_max (no duplicates), find_max (one test) — 660 is not 660 independent tasks but 33×20 generations with narrow checks.
3. 53M cannot absorb a 14B LLM: the student learns planning and family code; the general spectrum stays at runtime with the LLM (honest distillation limit).
4. Generation window of 150–300 tokens limits long programs.
5. No controlled substrate ablation at scale — open item.

7. Conclusion and Future Work

NeuroCA demonstrates the viability of the original thesis: the dynamics of a cellular automaton over registers can carry deterministic features under full state observability and honest exec verification. The hybrid "Transformer/MoE + CA substrate" reached 99.7% (658/660) on 33 algorithmic tasks; the series measured the role of capacity (G1), data (factor ~18× over the substrate at the current scale) and recorded valuable negative results (synthetic data, LoRA, head blocks, profile binarity before 53M).

Future work (order of application):

1. P0: fix the eval protocol (dedup, denominator by fact, functional Stack/Queue tests) before final measurements.
2. kit2c done (656/660, Tier1 485): edge data gave no gain — the hypothesis "Tier1 = uncovered inputs" is refuted; move to staged loss decomposition (task→analysis→strategy→code, loss masks) — curing the weak first stage without new parameters.
3. Distillation (path "b"): formulation diversity 33×3–5, Tier2 via the teacher (independent of V), student 53M on the distilled corpus.
4. Substrate ablation at ×5 data — separate "CA is useless" from "CA is useful at scale"; expert-specialization diagnostics (G4); experts 8→16 (G5).
5. Theoretical directions: hieroglyphic internal language, meta-observer, L-systems ("living text"), spatial understanding.

8. Reproducibility

8.1 Environment

Component | Specification

GPU | NVIDIA GeForce RTX 5080, 16 GB GDDR7 (Tier 0; CUDA)

CPU | AMD Ryzen 7 7840HS, 8C/16T Zen4 (Tier 1; exec harness, 16-thread pool)

iGPU | Radeon 780M (Tier 2; rendering/encoding only, never used for training)

Software | Python 3.12, PyTorch (bf16 autocast + GradScaler), NumPy, Ollama (HTTP 11434)

LLM teacher | qwen2.5:14b (semantics), qwen2.5-coder:14b / deepseek-coder (code), backup minimax-m3:cloud

Working root | D:\TOOLS_KA (paths below are absolute on the reference machine)

8.2 Artifacts (versioned)

Artifact | Path

Etalon checkpoint (53M) | D:\TOOLS_KA\orch_artifacts\hybrid_v120big50_kit2b_agent.pt

Etalon eval report | D:\TOOLS_KA\orch_artifacts\report_v120big50_kit2b.json

Feature cache (corpus) | D:\TOOLS_KA\orch_artifacts\branchA_kit2b.npz (1089 base-only rows), branchA_kit2c.npz (1161 rows)

Tier1 benchmark | D:\TOOLS_KA\orch_artifacts\bench_tier1.json (32 items)

Version registry | D:\TOOLS_KA\neuroca_lab\versions.json (26 entries, incl. per-task results and gates)

Eval logs (v114c/v114d/v115) | D:\TOOLS_KA\logs\v114c_eval_fixed_log.txt, v114d_eval_log.txt, v115_eval_fixed_log.txt

Teacher status | D:\TOOLS_KA\neuroca_lab\teacher_status.json, sched_queue.json

8.3 Commands

Training (Branch A, scratch; or INIT=... for transfer)

```
python D:/TOOLS_KA/train_branchA50.py branchA_kit2b.npz kit2b
```

Final 660 + Tier1 evaluation of a saved checkpoint

```
python D:/TOOLS_KA/eval_big50.py # 660 protocol
```

```
python D:/TOOLS_KA/eval_big50_t1.py # Tier1 protocol
```

Teacher pipeline (task generation → exec-verification → corpus)

```
python D:/TOOLS_KA/teacher_pipeline.py --tasks two_sum
```

Paper/PDF build (markers NEUROCA-TABLE in NeuroCA_article.md)

```
python D:/TOOLS_KA/make_article_pdf.py
```

8.4 Determinism and honest metrics

- Training seeds: ``torch.manual_seed(2024)``; per-sample eval seeds are explicit (300+k checks, 1000+k finals, 2000+ Tier1); per-run exec cache keyed by code hash.
- **Format guard is mandatory** for any eval: tests must be ``((args...), expected)``; the historical (a,b,expected) bug (v112) understated results (82.4% vs real 99.8%) — see §4.3.
- **Tier1 audit** (§6.4): ``res[fam]=got`` overwrites a repeated family (decimal_to_binary appears twice in bench_tier1.json); saved entries are $31 \times 20 = 620$, while the declared denominator is 640. The Tier1 figure must be read with this caveat; the fix is in progress.
- Stack/Queue families are scored by compile + presence of the class/function name (tests=None); functional tests are absent — see Appendix B.

8.5 License and data

Corpora (labels/*.jsonl, .npz feature caches) are project-internal; the code generated by the model is Python snippets derived from algorithmic task formulations. No personal data is involved. License for the preprint: CC BY 4.0 (suggested); for code/artifacts — to be declared by the laboratory.

Addendum (27.09.2026): root data defects and the new etalon

Two root defects found after this paper was written refine the results.

Corpus defect. `kit2_build.clean_text` silently dropped OOV words: 23,097 of 107,541 (21%) never reached training. The UNK fix (`NEUROCA_TOK_UNK=1` keeps the word position as UNK) restored the text; model `v120big50_unkbase` (transfer from `v121b`, seed 2024, 25 epochs) scored $660=479$ (old 345 / new 134) and $\text{Tier1}=355/620$ — $+45/+29$ over the previous etalon `kit2b` (434/326). Etalon updated; the 21% cut text explains part of the Tier1 edge deficits.

P2 format defect. None of the 2,094 training cues contained the "задача X: описание" construction that the P2 benchmark tests with — the model never saw the format, which explains the reversed bare/full-cue gap (373 vs 292). Treatment without using P2: 264 corpus descriptions converted to P2 format, 165 teacher paraphrases, textbook (29 tasks) in P2 format, vocab $V=2451$ (80% P2-word coverage), corpus now 279,042 pairs (+21.7%).

G7 substrate verdict. At 53M/104K the substrate features give no significant gain: scratch 332 ± 20 vs 366 ± 24 (−34, n.s.); transfer 434/326 vs 440/313 (noise). A model trained with the substrate, however, fails without it (cross-ablation 4/660): the substrate is a working support of the trained model. Branches L2/L3 are not built.

Eval protocol. The 658/660 (99.7%) figure was measured on the legacy bare-cue protocol ("задача X"). The strict full-cue protocol scores the same model 434/660, the new etalon 479/660: a stricter metric, not a regression.

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Appendix A. Etalon v120big50_kit2b details

Parameter | Value

Architecture | HybridPrefix d=256, L=6, H=8, MoE 8/3, R=2, FDIM=530

Parameters | 53M (Branch A)

Corpus | 1089 base-only rows (base 1054 + weak_fix 35), transfer from v121b

660 | 658/660 = 99.7% (old 460/460, new 198/200)

Tier1 | 487 over saved entries (31 families; 640 denominator — audit)

Weak base | is_balanced_parentheses 18/20

Tier1 = 0/20 | find_max, power, digit_sum, remove_spaces, count_digits

Tier1 drops | is_power_of_two 8, count_vowels 7, caesar_cipher 15

Checkpoint | orch_artifacts/hybrid_v120big50_kit2b_agent.pt

Report | orch_artifacts/report_v120big50_kit2b.json

Appendix B. Per-task breakdown 660/Tier1

Family		660		Tier1		Family		660		Tier1
factorial		20		20		average		20		20
fibonacci		20		20		second_max		20		20
is_palindrome		20		20		remove_spaces		20		0
find_max		20		0		is_leap		20		20
find_min		20		20		is_sorted		20		20
bubble_sort		20		20		count_digits		20		0
binary_search		20		20		is_power_of_two		20		8
is_prime		20		20		fizzbuzz		20		20
unique		20		20		is_anagram		20		19
count_vowels		20		7		caesar_cipher		20		15
count_words		20		20		is_balanced_parentheses		18		18
count_evens		20		20		decimal_to_binary		20		20
Stack		20*		—		merge_sorted		20		20
Queue		20*		—		longest_common_prefix		20		20
power		20		0		two_sum		20		20
gcd		20		20						
reverse_string		20		20						
digit_sum		20		0						

* Stack/Queue are scored by compile + presence of the class/function name (tests=None); no functional tests — a protocol limitation.

*Document compiled from the registry versions.json (26 versions), the article

NeuroCA_article.md, the plan PLAN.md (§12 "Reflections: Transformer on CA"),
the eval-protocol audit (26.09.2026) and scripts train_branchA50.py /
eval_big50_t1.py. All numerical values are direct copies from artifacts.*